

Griffin Downing PhD

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PROFESSIONAL SUMMARY: Life sciences professional with 10+ years of pre-clinical research experience. Eager about innovation in chronic illness and disease. Extensive experience leveraging scientific skills in metabolism and neuroscience research. Proven ability to effectively communicate complex scientific concepts to diverse audiences and a collaborative problem solver, as evidenced by a track record of interdisciplinary project completion in varied settings.

Skills:

Coding Languages and Statistical Tools: R Studio, Qualtrics, Prism, Excel.

Project Management: Collaboration, Problem Solving, Tracking of Deliverables, Managing stakeholders

Wet Lab: PCR, qPCR, ELISA, IHC/IF, In-Situ Hybridization, Westerns, Cell Culture and Microscopy (Confocal)

In-Vivo: Mouse Genetics, Surgery, Behavioral Assays, Pharmacology (PK/PD)

EDUCATION:

University of Michigan, Ann Arbor	2018 – 2024
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PhD in Molecular, Cellular, and Developmental Biology

San Francisco State University	2016 - 2018
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MS in Physiology and Behavioral Biology-ABT

- Awarded NIH Bridges to PhD Fellowship- \$18,000

University of Hartford - MS in Neuroscience	2014 - 2016
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University of California, Davis - BA in Psychology	2010 - 2013
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PROFESSIONAL EXPERIENCE:

Graduate Student Research Assistant, University of Michigan	2018 - 2024
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- Responsible for drafting, writing and reviewing protocols and standard safety procedures.
- Independently conducted research. Analyzed raw data and information using statistics and Microsoft Office, R Studio and Prism to present data as digestible.
- Managed a multidisciplinary team of students and scientists across multiple projects, fostering a collaborative research environment.
- Performed in vivo novel drug development and discovery testing in mouse models, resulting in a first author publication.
- Utilized single nucleus sequencing of genes to identify novel cellular markers involved in energy balance.
- Developed transgenic mice using molecular genetic techniques to investigate the underlying mechanisms of diseases such as obesity and anorexia, resulting in two publications
- Extensive experience in immunohistochemistry and imaging analysis techniques.

Consultant, miLEAD at the University of Michigan	2022 - 2024
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- Collaborated with clients to develop go-to-market strategies for successful product launches.
- Engaged with stakeholders to identify target populations, leveraging market segmentation insights to optimize marketing strategies.
- Led critical project components to support organizations in implementing innovative technologies, driving market adoption.
- Conducted comprehensive primary and secondary research to inform data driven recommendations and strategic decision making.

Graduate Student Researcher, San Francisco State University	2016 - 2018
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- Managed the laboratory with purchasing and organizing; implemented new approaches in the laboratory
- Utilized IHC/IF, Confocal Microscopy and ImageJ for IHC/IF analysis of tissues
- Initiated several collaborative projects in the lab with other departments (cell biology and genetics)
- Mentored graduate and undergraduate students in developing research skills and honing wet lab techniques; IHC/IF, ELISAS, Imaging, ImageJ analysis and statistics.

Medical Records Specialist/HIV Prevention Team, HGLHC**2014 - 2016**

- Led initiative for converting all paper documents to electronic medical records, streamlining documentation processes and improving data accessibility.
- Independently conducted research identifying target populations for HIV prevention and outreach
- Fostered team engagement to ensure quality of our HIV prevention and outreach efforts

Research Assistant/Junior Specialist, University of California, Davis**2010 - 2014**

- Managed and supervised other undergraduate students in the laboratory
- Conducted independent research and managed teams on projects, resulting in two co-author publications
- Use of immunohistochemistry and ImageJ to analyze c-Fos expression
- Rigorously trained in behavioral assays in used in rodent models

LEADERSHIP:

- **Graduate Student Mentor, University of Michigan** - Selected to be a mentor for new graduate student instructors and assist them with preparing for the challenges they might face **(2021)**
- **Outreach Coordinator, SACNAS at SFSU** - Director of social media outreach, maintained relationships with faculty and representatives from other universities for events **(2017 - 2018)**

SERVICE AND RECOGNITION:

- Volunteered with RAHMA, an organization based out of Washington, DC. Developed a toolkit for faith leaders to address the problem of female-genital mutilation in their communities. **(2020-2021)**
- Awarded NIH Bridges to PhD fellowship while at San Francisco State University **(2016-2018)**
- Invited speaker at Yale's Department of Evolutionary Anthropology. Talk Title: The Role of Oxytocin in Complex Social Behaviors in Prairie Voles **(2016)**

SELECT PUBLICATIONS:

- Wang S†, **Downing G†**, Olsen KF, Sawyer TK, Cone RD, Schwendeman SP. Aqueous remote loading of setmelanotide in poly (lactic-co-glycolic acid) microspheres for long-term obesity treatment. *J Control Release*. **(2023)**. <https://doi.org/10.1016/j.jconrel.2023.09.015>
† These authors contributed equally to this work.
- Gui Y, Dahir NS, Wu Y, **Downing G**, Sweeney P, Cone RD. Melanocortin-3 receptor expression in AgRP neurons is required for normal activation of the neurons in response to energy deficiency. *Cell Reports*. **(2023)** <https://doi.org/10.1016/j.celrep.2023.113188>
- Sweeney, P. Bedenbaugh, MN. Maldonado, J. Pan, P. Fowler, K. Williams, SY. Gimenez. LE. Ghamari-Langroudi, M. **Downing, GM**. Gui, Y. Hadley, CK. Joy, ST. Mapp, AK. Simerly, RB. Cone, RD. Melanocortin-3 receptor is a pharmacological target for the regulation of anorexia. *Science Translational Medicine*. **(2021)** <https://doi.org/10.1126/scitranslmed.abd6434>
- Guoynes, C.D., Simmons, T.C., **Downing, G.M.**, Jacob, S., Solomon, M., Bales, K.L. Chronic intranasal oxytocin has dose-dependent effects on central oxytocin and vasopressin systems in prairie voles (*Microtus ochrogaster*). *Neuroscience*, **(2018)** <https://doi.org/10.1016/j.neuroscience.2017.11.037>
- Bales KL, Perkeybile AM, Conley OG, Lee MH, Guoynes CD, **Downing GM**, Yun CR, Solomon M, Jacob S, Mendoza SP (2013). Chronic Intranasal Oxytocin Causes Long-Term Impairments in Partner Preference Formation in Male Prairie Voles. *Biological Psychiatry*, **(2013)** <http://doi.org/10.1016/j.biopsych.2012.08.025>